

Competitive Programming

Section 37: Segment Trees

Competitive Programming Course

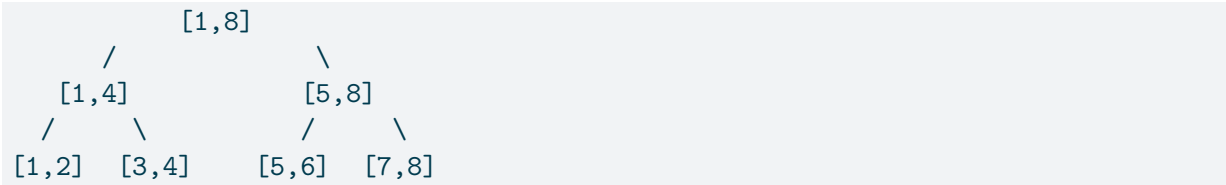
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1 Segment trees

A segment tree stores information about intervals. Each node owns a segment, and its value is computed by merging its two children.



If the merge is associative, such as `min`, `max`, `sum`, `gcd`, or a custom dynamic-programming state, a range query can be decomposed into $O(\log n)$ tree nodes.

2 Point update and range query

For point assignment, update one leaf, then recompute ancestors on the path to the root. This costs $O(\log n)$. Range minimum queries use `merge = min`.

Example with 1-based array:

```
a = [7, 2, 6, 3]
```

Leaves store:

```
[1,1]=7  [2,2]=2  [3,3]=6  [4,4]=3
```

Internal minimum nodes:

```
[1,2]=min(7,2)=2  
[3,4]=min(6,3)=3  
[1,4]=min(2,3)=2
```

If `a[2]` becomes 8, only nodes on the leaf-to-root path change:

```
[2,2], [1,2], [1,4]
```

This is why updates are logarithmic.

3 Rich node states

Segment trees are powerful because a node can store more than one number. For maximum subarray sum, each node stores:

```
sum    = total sum of the segment
pref   = best non-empty prefix sum
suff   = best non-empty suffix sum
best   = best non-empty subarray sum
```

To merge left L and right R:

```
sum    = L.sum + R.sum
pref   = max(L.pref, L.sum + R.pref)
suff   = max(R.suff, R.sum + L.suff)
best   = max(L.best, R.best, L.suff + R.pref)
```

This is a recurring red-level idea: design exactly the information that makes two halves composable.

4 Lazy propagation

If an update covers a whole node segment, do not immediately walk into all leaves. Store a pending tag on the node. For range-add and range-sum:

```
seg[node] += delta * segment_length  
lazy[node] += delta
```

Before visiting children, push the tag to them. Lazy propagation turns range update plus range query from $O(n)$ to $O(\log n)$.

5 Why the maximum-subarray state works

When two neighboring segments are merged, the best subarray in the union is one of exactly three types:

```
entirely in left  
entirely in right  
crosses the boundary
```

The crossing case must be:

```
best suffix of left + best prefix of right
```

That is why the node stores `pref` and `suff` in addition to `best`.

6 Lazy propagation example

Suppose a node covers $[5,8]$ and stores sum 20. Adding +3 to the whole range changes the sum by:

```
4 positions * 3 = 12  
new sum = 32
```

We record `lazy += 3` and stop. If a later query needs $[5,6]$, we push the tag to children before descending:

```
left child [5,6] gets +3 on 2 positions  
right child [7,8] gets +3 on 2 positions
```

The parent tag is then cleared.

7 Mistakes to avoid

Mixing inclusive and half-open ranges.

Forgetting that maximum subarray is non-empty in this section.

Applying lazy delta to `seg[node]` without multiplying by segment length.

Not pushing lazy tags before descending into a partially covered node.

8 Local exercises

- A. Range Minimum Updates: Point-assign values and answer range minimum queries.
- B. Maximum Subarray Updates: After each point assignment, print the maximum non-empty subarray sum of the whole array.
- C. Lazy Range Add Sum: Support range add updates and range sum queries.

9 How to use the checker

The local checker compiles/runs both languages and compares output tokens exactly:

```
CP_TARGET=solution python3 sections/37_segment_trees/check.py
```

Replace `CP_TARGET=solution` by your own `solve.cpp` or `solve.py` attempts when practicing.
Exercises

Use A to define a range-minimum merge, B to include prefix/suffix/best values in a merge, and C to state a lazy tag before coding. Test an update on one element, a full-range update, and tag composition order.

10 Choosing and proving a segment tree

Use a segment tree when updates invalidate precomputed answers and the requested range summary can be merged from two adjacent summaries. The node invariant is:

```
node v stores the exact summary of its own interval.
```

For a point assignment, only the leaf and its ancestors change. For a range query, the iterative or recursive walk selects disjoint canonical intervals whose union is the query range. Associativity of the merge makes their combined summary equal the answer.

10.1 Trace: iterative range minimum

For $a=[5,4,3,2,1]$, change position 3 to 10. The leaf becomes 10; recomputing its parent, grandparent, and root restores the invariant. Query $[2,4]$ sees the values $[4,10,2]$, so its minimum is 2. In the half-open iterative form, translate 1-based inclusive $[l,r]$ to leaf pointers $[l-1+size, r+size)$. The right endpoint is deliberately exclusive.

10.2 Trace: maximum subarray merge

Merge left $[1,-2]$ and right $[3,4]$. Left state is $sum=-1, pref=1, suff=-1, best=1$; right state is $sum=7, pref=7, suff=7, best=7$. The crossing candidate is $-1+7=6$, so the best is 7. The crossing candidate is always $left.suff+right.pref$; no other crossing subarray is possible.

10.3 Lazy trace

If a node for $[1,4]$ has sum 10 and receives $+3$, its summary becomes 22 and its pending tag becomes 3. Do not descend. If a later request partially touches the node, push: both length-two children gain $3*2$ in their sums and gain a pending $+3$. Pushing preserves the parent summary while making child summaries truthful enough to recurse.

11 Implementation choices and failure modes

Pad the tree to a power of two. For unused leaves of a minimum tree use `INF`; for non-empty maximum-subarray trees use neutral (`sum=0, pref=suff=best=-INF`), never all zeros. For lazy range-add/range-sum, `apply(v, len, x)` must change the sum by `x*len`, and tags compose by addition. Use `long long` in C++ and ordinary Python integers for all sums.

Mistake	Revealing test
use zero as min padding	all array values positive
allow empty subarray	all values negative
forget length in lazy apply	full-range update then sum query
descend without push	full update, then partial query
mix <code>[l,r]</code> and <code>[l,r)</code>	one-element query

11.1 Formative checks

1. Why does `best` need `suff` and `pref`? Answer: a best subarray crossing a merge boundary consists exactly of the left suffix plus the right prefix.
2. What does a lazy tag mean? Answer: every element in the node's interval has received that unapplied delta, while `seg[v]` already includes its effect.
3. What is the answer after assigning `-5` to every element of `[-1,-2]`? Answer: `-5`; the section requires a non-empty subarray.

12 Submission workflow

Stress a tree against a literal array for short random sequences. Include overlapping lazy updates, negative deltas, assignment of the same position many times, and queries at both boundaries. Run:

```
CP_TARGET=solution python3 sections/37_segment_trees/check.py
```